

Question Paper

Exam Date & Time: 23-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

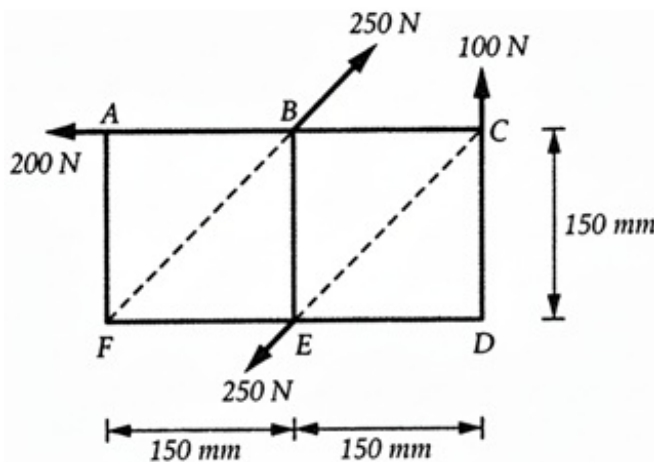
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 25/JAN 26
SUBJECT: CIE 1071/CIE_1071 MECHANICS OF SOLIDS

Marks: 50

Duration: 180 mins.

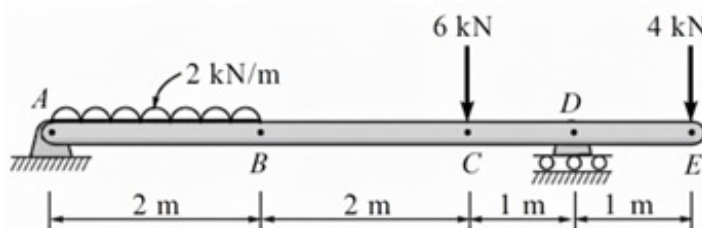
Answer all the questions.

- 1A) A rectangular plate of dimensions $150\text{ mm} \times 300\text{ mm}$ is subjected to four forces acting at various points as shown in the figure. Determine magnitude and direction of the resultant force and its location from edge C of the plate. (5)



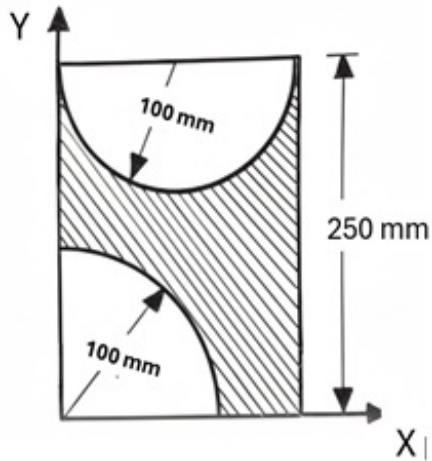
Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Determine the reactions at all the supports of the beam shown in the figure below. (5)



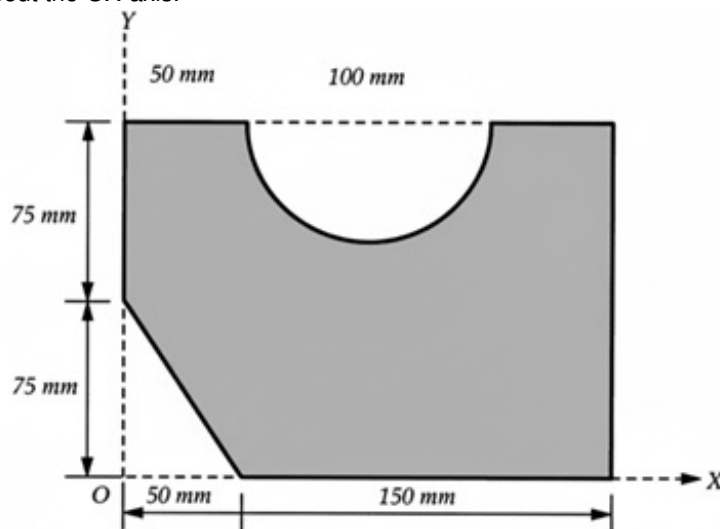
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Determine the centroid for the shaded portion shown in the figure w.r.t axes shown. (5)



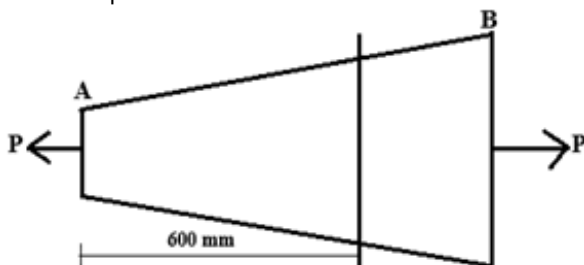
Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) For the plane area shown in the figure, determine the moment of inertia and the radius of gyration about the OX-axis. (5)



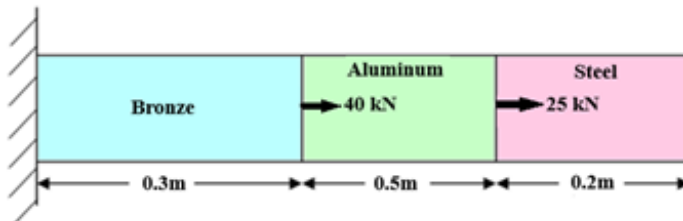
Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) A tapered bar of rectangular cross section is 40 mm wide at one end and 75 mm wide at the other, 10 mm thick and 1 m long. The elongation of 0.1 mm was observed under load P. Determine the load P, if the modulus of elasticity of the material of the bar is 100 GPa. Also calculate stress at 600 mm from point A. (5)



Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) A composite prismatic bar consists of three sections rigidly fastened in series as shown in the figure below. Determine the stress in each section, change in length of each section and total change in length of the bar. (5)



	bronze	aluminum	steel
E (GPa)	100	70	200
A (mm²)	2500	3000	2000
L (m)	0.3	0.5	0.2

Course Outcome: "CO4" / Cognitive Level: "Application"

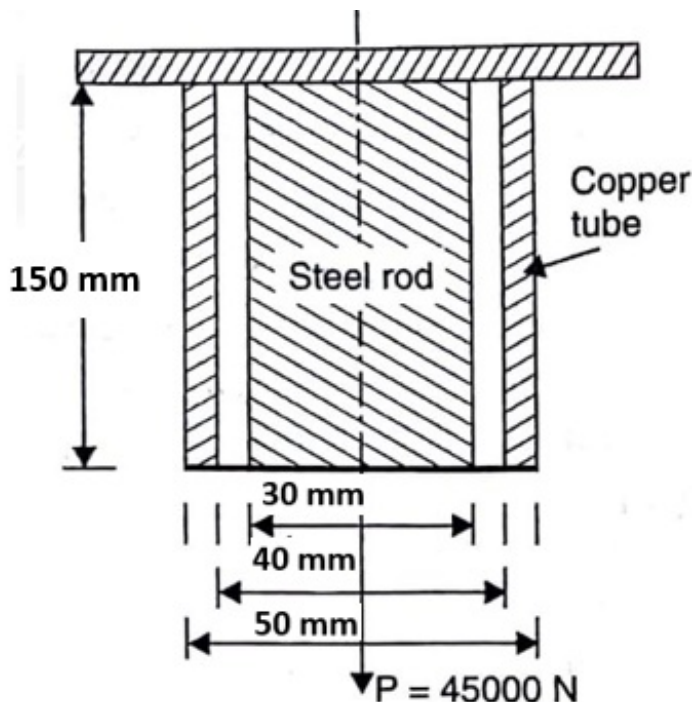
- 4A) A circular rod of 30 mm diameter and 550 mm long is subjected to a tensile force of 70 kN. (5)
Determine modulus of rigidity, bulk modulus and change in volume if Poisson's ratio = 0.3 and Young's modulus $E = 2 \times 10^5 \text{ N/mm}^2$.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) A thin cylindrical shell is 4m long and 1m in internal diameter. It is subjected to an internal pressure of 1.4 MPa. If the thickness of the sheet is 14mm, determine the circumferential stress, longitudinal stress, changes in diameter, length and volume. Take $E=200 \text{ GPa}$ and $\mu=0.3$. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) A steel rod of 30 mm diameter is enclosed centrally in a hollow copper tube of external diameter 50 mm and internal diameter of 40 mm. The composite bar is then subjected to an axial pull of 45 kN. If the length of each bar is equal to 150 mm, determine: (i) The stresses in the rod and tube, and (ii) The load carried by each bar. Take E for steel $=2.1 \times 10^5 \text{ N/mm}^2$ and for copper $=1.1 \times 10^5 \text{ N/mm}^2$. (5)

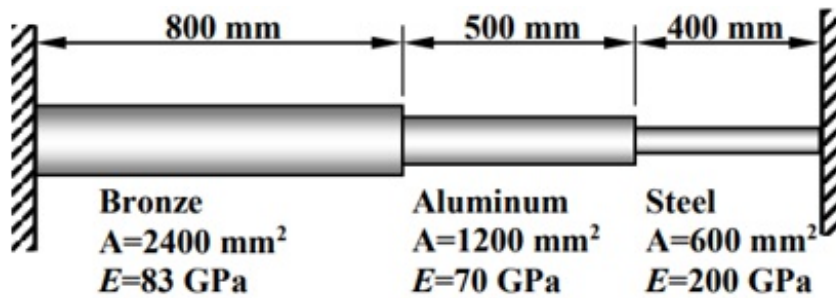


Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) A rod is composed of three segments, as shown in the figure. Compute the stress induced in each (5)

material by a temperature drop 30 °C if (a) the walls are rigid and
(b) Both the supports yield by 0.3 mm.

Take $E_s = 200 \text{ GPa}$, $E_b = 83 \text{ GPa}$, $E_a = 70 \text{ GPa}$, $\alpha_s = 11.7 \times 10^{-6}/^\circ\text{C}$,
 $\alpha_b = 18.9 \times 10^{-6}/^\circ\text{C}$,
 $\alpha_a = 23 \times 10^{-6}/^\circ\text{C}$.



Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CIE_1071:MECHANICS OF SOLIDS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	20
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CIE_1071:MECHANICS OF SOLIDS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)